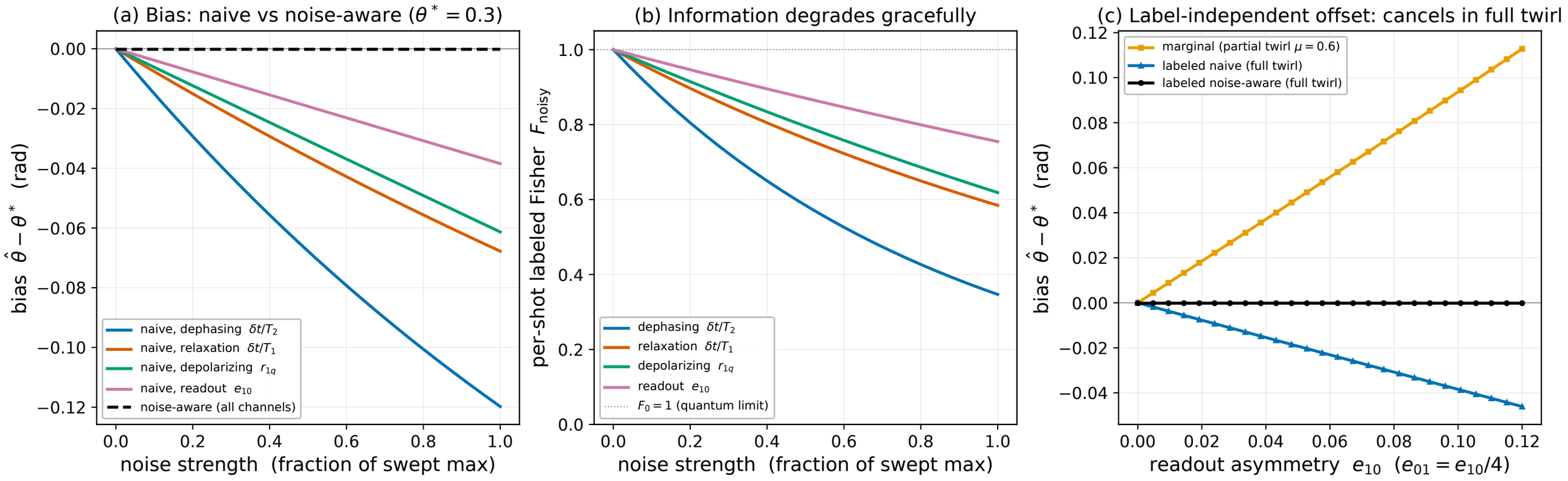
(a) Bias: naive vs noise-aware ($\theta^* = 0.3$)
bias $\hat{\theta} - \theta^*$ (rad)
noise strength (fraction of swept max)
naive, dephasing $\delta t/T_2$
naive, relaxation $\delta t/T_1$
naive, depolarizing r_{1q}
naive, readout e_{10}
noise-aware (all channels)
(b) Information degrades gracefully
per-shot labeled Fisher F_{noisy}
noise strength (fraction of swept max)
dephasing $\delta t/T_2$
relaxation $\delta t/T_1$
depolarizing r_{1q}
readout e_{10}
$F_0 = 1$ (quantum limit)
(c) Label-independent offset: cancels in full twirl
bias $\hat{\theta} - \theta^*$ (rad)
readout asymmetry e_{10} ($e_{01} = e_{10}/4$)
marginal (partial twirl $\mu = 0.6$)
labeled naive (full twirl)
labeled noise-aware (full twirl)